Improving Multi-Class Mental Health Text Classification with BERT: A Comparison of Model Optimization Techniques with a Focus on Minority Class Detection

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1. Abstract:

Mental health conditions affect hundreds of millions of people worldwide, yet proactive and accurate detection remains a significant challenge. This study investigates how hyperparameter tuning, targeted data augmentation, and domain-specific architecture compare in improving BERT-based multi-class mental health text classification across seven mental health categories. Using a publicly available social media dataset, experiments were conducted in sequential stages with macro F1 and evaluation accuracy as the primary evaluation metrics. A strong baseline BERT-base model achieved a macro F1 of 83.34% and accuracy of 84.46%. Hyperparameter tuning yielded modest gains, given that the baseline hyperparameters were so close to the optimal configuration. Targeted augmentation of the Stress class produced the most substantial improvements. MentalBERT, combined with augmentation and optimized hyperparameters, achieved the highest performance with a macro F1 of 86.25% and accuracy of 86.19%. Persistent misclassification between Depression and Suicidal remained unresolved throughout. Results demonstrate that the strongest performance emerged from combining all three strategies rather than relying on any single approach.

Keywords:

BERT; MentalBERT; Hyperparameter tuning; Data augmentation; Macro F1; Minority class detection

2. Introduction:

Mental health issues such as depression, anxiety, and stress are increasingly recognized as critical public health concerns, affecting hundreds of millions of people worldwide. Modern society–with its rising academic, economic, and social pressures–only contributes to the exacerbation of mental health challenges, leaving individuals in a heightened state of stress, isolation, and vulnerability. In today's digital age, people increasingly share their feelings and struggles through online posts, messages, and social media updates. This vast amount of text holds valuable clues that, if analyzed correctly, could enable proactive intervention and support for those at risk.

However, classifying mental health conditions from text data poses several challenges. Datasets in this domain are often highly imbalanced, where common conditions like depression are abundant, while others, such as personality disorders, have far fewer samples. This imbalance leads to biased models that perform well on frequent classes but poorly on minority ones. Additionally, standard transformer architectures like BERT, while powerful, are limited in tasks like these where domain-specific pre-training is often required. Fortunately, advances such as domain-adaptive models such as MentalBERT and class-specific data augmentation techniques offer promising avenues to mitigate these challenges and improve both overall and minority class detection.

The long-term objective of this research is to contribute to early intervention and scalable mental health screening across public platforms, ultimately improving

access to care and well-being for individuals at risk. This research project offers practical insights and guidance for researchers working on similar tasks, developers building mental health classifiers for short texts, and practitioners managing challenges like class imbalance and hyperparameter tuning in this specific domain. In pursuit of this goal, this project explored a series of modeling strategies—including hyperparameter tuning, targeted data augmentation, and the use of domain-specific transformer architectures—to improve BERT-based models' ability to classify mental health text. The aim was to maximize macro-F1 scores through enhancing detection of both common and less frequent mental health classes. The project objectives were as follows:

Research Question:

How do the modeling strategies hyperparameter tuning, class-specific data augmentation, and domain-specific architecture compare in maximizing BERT's performance in multi-class mental health text classification, with an emphasis on improving minority class detection?

Objectives:

1. Run a hyperparameter grid search on BERT-base and MentalBERT to identify optimal combinations of learning rate, batch size, and maximum sequence length for maximizing macro F1 score and accuracy.
2. Evaluate the impact of class-specific data augmentation techniques on model performance, particularly for the underrepresented classes Stress and Personality Disorder.
3. Compare the effectiveness of BERT-base with more specialized architectures such as MentalBERT when combined with class-targeted augmentation and tuning, and determine whether these strategies support stronger performance.
4. Determine which modeling strategies work best to improve BERT's capacity in multi-class mental health classification. Provide practical insights for building NLP-based tools that can assist in early identification of mental health concerns.

3. Literature Review

In recent years, natural language processing (NLP) has achieved state of the art performance in various tasks including text classification and sentiment analysis. Specifically, some models have been used to analyze mental health signals in non-clinical texts such as social media posts. Early work by Calvo et al. (2017) highlighted the promise of extracting meaningful mental health insights from such user-generated data, while also underscoring the challenges posed by its informal, brief, and emotionally complex nature. These characteristics make mental health text classification both valuable and methodologically complex.

Advancements in transformer architectures, particularly models like BERT, have significantly improved the capacity of models to perform semantic analysis tasks (Martínez-Castaño et al., 2021; Sao & Lim, 2024). Unlike a recurrent neural network (RNN), an earlier approach to NLP, transformer architectures incorporate attention mechanisms which weight individual input elements, enabling them to detect nuanced contextual representations. This makes transformers like BERT particularly advantageous for mental health applications, where subtle linguistic cues are essential indicators of psychological distress. MentalBERT, introduced by Ji et al. (2021), is a domain-adapted variant of BERT, fine-tuned specifically on mental health-related Reddit text data. It outperforms general-purpose BERT models in both binary and multi-class mental health classification tasks, offering a stronger alternative to the standard BERT base model.

Despite promising results, significant challenges remain. A key concern is the imbalance between well-represented mental health conditions and rarer ones, which often leads to models biased toward majority classes (Chancellor & De Choudhury, 2020). Studies like those by Sao and Lim (2024) and Martínez-Castaño et al. (2021)

show that hyperparameter choices, choices that dictate the training process for a model, can influence not just overall performance but also the model's sensitivity to less frequent categories. Work by Ansari et al. (2021) explored the effect of data augmentation techniques such as synonym replacement and backtranslation on BERT-based mental health text classifiers. Results showed "significant improvements in classifiers performance when trained on the augmented data" (Ansari et al). Through effective data augmentation, a more balanced dataset is created, one that provides sufficient data for BERT.

Research by Guo et al. (2022) indicates that the effectiveness of BERT-based models is inextricably tied to careful hyperparameter tuning, including learning rate, batch size, and sequence length, to ensure high accuracy and to manage computational demands. Their comparative study of pretraining strategies emphasizes that even well-constructed models may underperform if not fine-tuned appropriately to the domain-specific nature of social media mental health data. These findings suggest that optimal performance is not just a matter of model selection, but of strategic training, where hyperparameter choices play a pivotal role in balancing generalization and overfitting, especially in tasks involving imbalanced datasets and varied text lengths.

4. Methodology

Dataset and Preprocessing

The dataset used in this project was obtained from Kaggle. The dataset (Table 1) is a collection of short text statements organized under the following mental health categories: *Normal, Depression, Anxiety, Stress, Suicidal, Bipolar*, and *Personality Disorder*. The text data was collected from various public forums on social media platforms such as Twitter and Reddit.

Table 1. Sample of raw data.

	text	label
0	oh my gosh	Anxiety
1	trouble sleeping, confused mind, restless hear...	Anxiety
2	All wrong, back off dear, forward doubt. Stay ...	Anxiety
3	I've shifted my focus to something else but I'...	Anxiety
4	I'm restless and restless, it's been a month n...	Anxiety

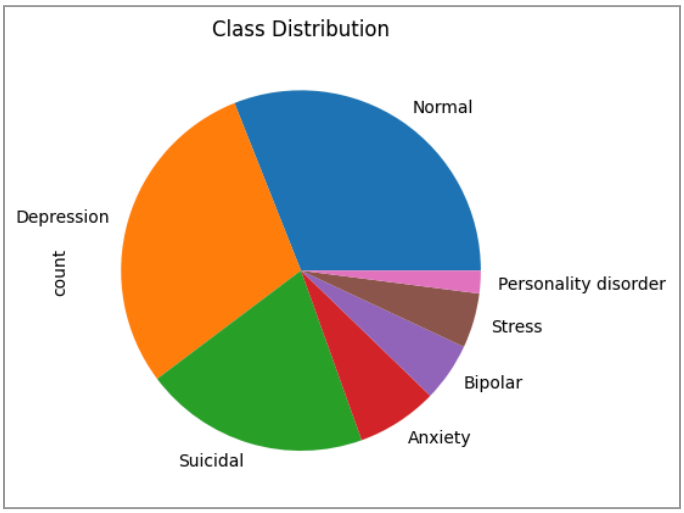

Figure 1. Class distribution of mental health text dataset.

All rows with missing classifications and blank text statements were effectively eliminated from the dataset. Following, the dataset was stratified using a 80-20 train-test split. All text samples were preprocessed using the BERT tokenizer and padded/truncated to a maximum sequence length of 256 tokens to ensure consistency across inputs. The dataset exhibited a highly imbalanced class distribution with the majority of samples being attributed to more commonly seen categories such as *Normal* and *Depression* while minority classes such as *Bipolar*, *Stress*, and *Personality Disorder* were severely underrepresented (See Figure 1).

Evaluation Metrics

Performance will be assessed via:

1. *Macro F1 score:* Evaluates model performance equally across all classes, ensuring balanced representation
2. *Test accuracy:* Measures overall accuracy of predictions; inherently weighted by class

frequency, providing a complementary view to macro F1 scores

3. *Confusion matrices:* Visualize misclassifications to identify patterns
4. *Per-class F1 Score:* Tracks performance on individual classes, highlighting progress for minority classes

Experimental Setup

Experiments were conducted using PyTorch and the HuggingFace Transformers library in a Google Colab workspace, with GPU acceleration enabled when available. Random seeds were set for reproducibility of training and evaluation results. For the baseline run, text data was tokenized using the BERT tokenizer and truncated/padded to a maximum sequence length of 256 tokens. Training and test datasets were loaded into PyTorch dataloaders with batch sizes of 64. This process was repeated using different values during experiment runs.

BERT for sequence classification (`bert-base-uncased`) was initialized with a single linear layer on top of the pretrained base model. The AdamW optimizer ($\varepsilon = 1e\text{-}8$) was used with a baseline learning rate of 0.0001 alongside a linear scheduler with no warmup steps. Preliminary tests comparing two, three, and four epochs showed that two epochs best balanced computational capacity and effective convergence. Additionally, training loss consistently plateaued at around 0.13 in the fourth epoch, with little to no improvement in model performance thereafter. As a result, models were trained for two epochs. Shown below is *Table 2*: results from one of the preliminary tests which culminated in the decision to train for two epochs.

Table 2. Results show that training for more than 2 epochs produces minimal gains (if any).

Epoch	Accuracy	Macro F1	Avg Train Loss
2	83.499	82.251	0.3480
3	83.382	82.237	0.2249
4	83.354	82.539	0.1305

Custom training and evaluation loops were set up to streamline the model running process. The training loop processed batches of tokenized inputs on the GPU. Model weights were updated using the AdamW optimizer with gradient clipping, and a linear learning rate scheduler. The training objective was standard cross-entropy loss for multi-class classification. Average training loss was recorded each epoch with intermittent updates every 300 steps. The evaluation loop was used to generate predictions, and then calculate and return metrics to assess performance across both common and minority classes. These custom loops assisted in model running and the conducting of grid searches.

5. Experimentation and Results

The experiments were structured in stages, with each stage exploring a specific modeling approach and its impact on classification performance. The primary goal was to compare the effectiveness of different modeling strategies (hyperparameter optimization, targeted data augmentation, and domain-specific architectures) and to identify the combination that maximized overall model performance, particularly in terms of macro F1 score and accuracy. Each stage introduced controlled adjustments to hyperparameters such as learning rate, batch size, and sequence length while keeping other variables constant to ensure fair comparison.

Stage 0: Baseline Modeling

The baseline BERT-base model (`bert-base-uncased`) was trained with a learning rate of 0.0001, a batch size of 64, and a maximum sequence length of 256.

Overall Performance:

Accuracy: 84.46%

Macro F1 Score: 83.34%

Average Training Loss: 0.3212

Figure 2. Baseline Model training loss graph over 2 epochs

Table 3. Baseline Model confusion matrix

	Pred 0	Pred 1	Pred 2	Pred 3	Pred 4	Pred 5	Pred 6
0. Anxiety	679	18	18	10	9	20	1
1. Bipolar	14	459	30	6	5	13	0
2. Depression	19	16	2345	40	4	14	578
3. Normal	15	1	24	3177	2	64	25
4. Personality Disorder	12	4	26	6	171	18	0
5. Stress	30	7	28	40	7	423	1
6. Suicidal	1	2	483	25	0	2	1645

Observations:

- Larger classes such as *Normal* achieved strong performance (F1 ≈ 0.961), whereas other large classes like *Depression* and *Suicidal* were commonly misclassified with each other, resulting in lower F1 scores (0.786 and 0.746, respectively).
- Minority classes, particularly *Personality Disorder*, exhibited lower recall, yielding an F1 score of 0.786 despite reasonable precision.
- The macro F1 score being slightly lower than overall accuracy indicates that class imbalance negatively affected minority class detection.

Stage 1: Hyperparameter Tuning on BERT-base

Hypothesis 1.1: A learning rate of 5e-5 will improve macro F1 performance by allowing the model to learn subtle patterns in underrepresented classes without overshooting the loss minimum.

Hypothesis 1.2: A moderate batch size of 32 will reduce noise while improving F1 scores for underrepresented classes and keeping training time manageable on a large dataset, leading to higher macro F1 scores.

Hypothesis 1.3: Given the observed text length average of 113.3 tokens, a maximum sequence length of 128 tokens will provide sufficient context for classification without increasing the risk of overfitting, resulting in better macro F1 performance.

Following the baseline evaluation, hyperparameter tuning was conducted in the form of a grid search with learning rates of [1e-5, 5e-5, 1e-4]; batch sizes of [16, 32, 64]; and maximum sequence lengths of [128, 256].

Table 4.1 Macro F1 Scores (%)

max_len	batch_size	lr=0.00001	lr=0.00005	lr=0.0001
128	16	82.047	83.486	81.975
128	32	77.436	81.961	81.939
128	64	77.058	82.251	81.778
256	16	80.162	83.342	82.984
256	32	80.265	84.055	83.356
256	64	81.476	82.899	83.337

Table 4.2 Evaluation Accuracy (%)

max_len	batch_size	lr=0.00001	lr=0.00005	lr=0.0001
128	16	82.624	82.75	82.965
128	32	81.731	83.354	83.202
128	64	80.907	83.449	83.373
256	16	83.781	84.692	84.593
256	32	83.156	84.806	84.626
256	64	84.212	84.597	84.455

Table 4.3 Average Training Loss

max_len	batch_size	lr=0.00001	lr=0.00005	lr=0.0001
128	16	0.3694	0.3219	0.3565
128	32	0.4561	0.3268	0.3514

128	64	0.4608	0.348	0.3339
256	16	0.3651	0.317	0.3159
256	32	0.4101	0.3058	0.3136
256	64	0.3492	0.3303	0.3212

The grid search across learning rate, batch size, and maximum sequence length showed that a learning rate of 5e-5, batch size of 32, and sequence length of 256 yielded the best overall performance. The tuned model achieved an accuracy of 84.806% and a macro F1 score of 84.055%, marking a modest improvement over the baseline. All classes showed slight gains in F1 score, though *Personality Disorder* remained limited by low recall, and confusion between *Depression* and *Suicidal* persisted. These results suggest that while hyperparameter tuning improved stability and generalization, further model enhancement required data-level intervention and/or architectural modification.

Stage 2: Targeted Data Augmentation

Hypothesis 2.1: When hyperparameters are kept consistent, applying data augmentation to class 4 (*Personality Disorder*) and class 5 (*Stress*) will yield a higher Macro F1 Score and Evaluation Accuracy than the "Tuned BERT-base" model.

To address persistent class imbalance, synonym replacement was performed using the NLPaug library. The augmentation nearly doubled the sizes of the 2 smallest classes: *Personality Disorder* and *Stress*. In Model "Augmented V1," (the first model iteration leveraging data augmentation) both minority classes were expanded using synonym replacement.

However, while *Stress* improved from an F1 score of 78.93% to 81.20%, *Personality Disorder* unexpectedly dropped from 79.82% to 61.16%. Correspondingly, macro F1 score fell to 81.62%. In Model "Augmented V2," only the *Stress* class was augmented while keeping all hyperparameters consistent (learning rate = 5e-5, batch size = 32, max sequence length = 256). This version achieved a stronger overall performance than both "Augmented V1" and "Tuned BERT-base," with 85.60% accuracy and a macro F1 score of 85.42%. The macro F1 gains primarily came from improved performance on underrepresented classes. Notably, the F1 score for the *Stress* class jumped up to 87.43% surpassing that of "Augmented V1" by over 6.2%. *Personality Disorder* also recovered and exceeded its previous F1 scores with a score of 80.49%. Other classes showed slight but consistent improvements.

Table 5. Augmented V2 confusion matrix

	Pred 0	Pred 1	Pred 2	Pred 3	Pred 4	Pred 5	Pred 6
0. Anxiety	695	17	17	11	8	17	3
1. Bipolar	17	482	27	4	7	17	2
2. Depression	10	24	2398	35	2	8	604
3. Normal	16	1	28	3148	4	50	22
4. Personality Disorder	5	5	23	8	165	9	0
5. Stress	27	6	17	54	9	748	1
6. Suicidal	2	0	420	30	0	0	1678

748 statements from the *Stress* class were predicted correctly, marking a significant improvement over the 423 *Stress* class statements from the baseline BERT-base model.

Stage 3: Domain-Specific Architecture

Hypothesis 3.1: MentalBERT's advanced and domain-specific architecture will outperform the "BERT-base baseline" model, primarily in minority class detection.

Hypothesis 3.2: The data augmented MentalBERT model will outperform the baseline MentalBERT model when only class 5 (*Stress*) is augmented.

To further improve classification performance, the model architecture was changed from the original BERT-base to MentalBERT (`mental/mental-bert-base-uncased`). MentalBERT is a domain-specific BERT model pre-trained on mental

health–related texts, such as Reddit posts from mental health forums. Unlike the original BERT, which is trained on general-purpose text like Wikipedia and BooksCorpus, MentalBERT is fine-tuned to better understand the linguistic patterns, emotional cues, and informal expressions common in mental health discussions.

To ensure fair comparison with previous experiments, the MentalBERT baseline was trained for 2 epochs on the non-augmented dataset using the same hyperparameters: a learning rate of 5e-5, a batch size of 32, and a maximum sequence length of 256. This baseline MentalBERT model achieved an accuracy of 85.08% and a macro F1 score of 83.80%. These results were slightly lower than expected, demonstrating the effect of class imbalance. The individual F1 scores were similar to those of the "Tuned BERT-base" model, indicating potential for class-specific data augmentation.

Next, the targeted data augmentation strategy from Stage 2 was applied to MentalBERT. Specifically, synonym replacement was used to augment the *Stress* class, which had previously shown strong responsiveness to augmentation.

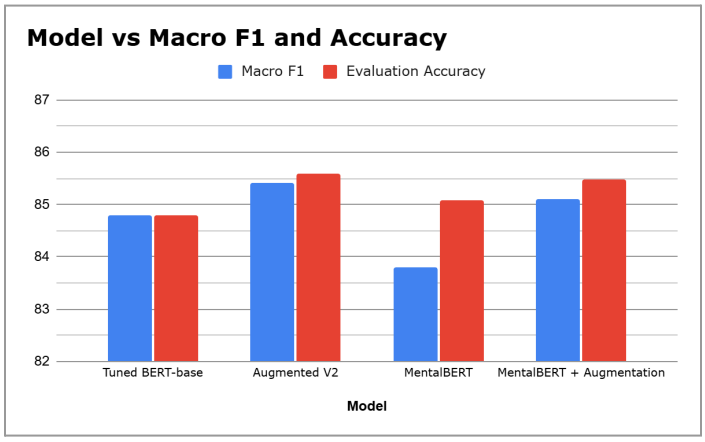

Figure 3. Graph of Model vs Macro F1 and Accuracy

The augmented MentalBERT model achieved an accuracy of 85.49% and a macro F1 score of 85.11%. These results showed clear improvement over the MentalBERT baseline model in regards to performance on underrepresented classes. However, overall performance remained slightly below that of "Augmented V2." Augmenting the *Stress* class improved its F1 score across both architectures. Additionally, *Depression* and *Personality Disorder* exhibited similar F1 scores despite large differences in class size. This is primarily attributed to frequent misclassification between the *Depression* and *Suicidal* classes, highlighting the semantic overlap between these categories.

To fully leverage the MentalBERT architecture, a grid search was conducted on the augmented dataset with learning rates of [5e-5, 1e-4]; batch sizes of [16, 32, 64]; and a maximum sequence length 256. This resulted in a total of six possible hyperparameter configurations.

Table 6.1 Macro F1 Score (%)

batch_size	lr=0.00005	lr=0.0001
16	86.202	85.437
32	85.111	86.261
64	86.041	86.346

Table 6.2 Evaluation Accuracy (%)

batch_size	lr=0.00005	lr=0.0001
16	86.159	85.433
32	85.488	86.02
64	86.104	86.123

Table 6.3 Average Training Loss

batch_size	lr=0.00005	lr=0.0001
16	0.2775	0.3191
32	0.3015	0.2816
64	0.2857	0.2746

The optimal configuration produced an accuracy of 86.12% and a macro F1 score of 86.35%. This also marked the highest performance across all stages of the research study.

6. Discussion

6.1 Performance Progression

This study evaluated and compared the impact of hyperparameter tuning, targeted data augmentation, and domain-specific pretraining on transformer-based multi-class mental health classification. Performance improved consistently across experimental stages. The baseline BERT model achieved:

Macro F1 of 83.34%

Accuracy of 84.46%

The final optimized MentalBERT model achieved:

Macro F1 of 86.25%

Accuracy of 86.19%.

A 2.91% gain in macro F1 and a 1.73% increase in accuracy may appear modest. However, these performance gains reflect meaningful improvement and balanced classification across all classes, especially underrepresented categories. By the conclusion of the experimental stage, minority class gains were the most pronounced.

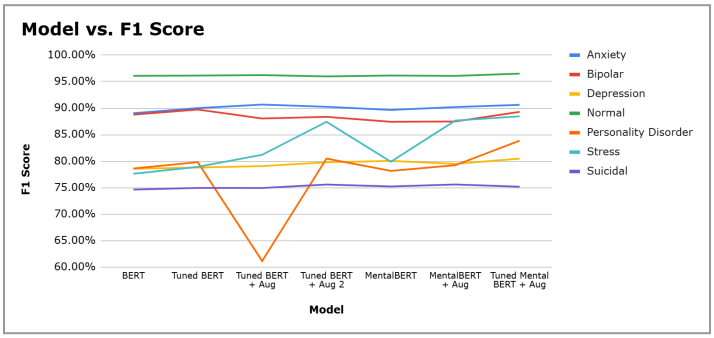

Figure 4. Graph of Model vs individual F1 scores

6.2 Strategy Comparison and Minority Class Detection

6.2.1 Hyperparameter Tuning

Hyperparameter tuning modestly improved macro F1 score. Tuning BERT-base yielded a rise in macro F1 from 83.34% to 84.06%. Similarly, hyperparameter tuning MentalBERT on the augmented dataset increased macro F1 from 85.11% to 86.25%. Improvements were consistent across all classes, indicating training stability and efficient convergence. The magnitude of change was limited. However, it is important to keep in mind that the hyperparameter grid searches revealed a wide range of performances. The chosen baseline coincidentally happened to be on the higher end of the scale, with hyperparameters close to the optimal configuration.

Hyperparameter tuning did not solve many of the persistent problems such as imbalanced performance and distinguishing between similar classes. This indicates that hyperparameter tuning primarily affects optimization dynamics rather than the model's true ability to semantically distinguish between mental health conditions.

6.2.2 Targeted Data Augmentation

Targeted data augmentation produced the most substantial performance gains, but demonstrated sensitivity to implementation strategy. When applied to both the *Personality Disorder* and *Stress* classes, augmentation significantly degraded *Personality Disorder* performance: F1 score dropped from 79.82% to 61.16%. Given the small size of the *Personality Disorder* class, data augmentation likely introduced more augmented statements than not. This would have created noise for the model and distracted it from learning the features and attributes of the *Personality Disorder* class. In contrast, selective augmentation applied only to the *Stress* class led to drastic improvement. Augmenting only the *Stress* class improved its F1 score by 8.5% with BERT-base and 7.8% with MentalBERT over each model's baseline. These results highlight the strong potential of targeted data augmentation for improving minority class detection.

6.2.3 Domain-Specific Architecture

MentalBERT, pretrained on mental health-related text, achieved a baseline macro F1 of 83.80 and an accuracy of 85.081. The macro F1 score was lower than the accuracy, indicating that domain-specific pretraining alone did not solve the issue of class imbalance. However, when combined with selective augmentation and optimized hyperparameters, MentalBERT achieved the highest overall

performance: macro F1 score of 86.25%. This suggests that domain-specific architectures provide measurable benefits when integrated with supporting optimization strategies.

6.3 Evaluation of Hypotheses

Hypothesis 1.1: A learning rate of 5e-5 will improve macro F1 performance by allowing the model to learn subtle patterns in underrepresented classes without overshooting the loss minimum.

- Evaluation: Supported. The learning rate stabilized training and improved minority class detection better than lr = 1e-4 and lr = 1e-5.

Hypothesis 1.2: A moderate batch size of 32 will reduce noise while improving F1 scores for underrepresented classes and keeping training time manageable on a large dataset, leading to higher macro F1 scores.

- Evaluation: Supported. Batch size 32, only when coupled with maximum sequence length of 256 provided stable training and moderate improvements in minority class F1 scores.

Hypothesis 1.3: Given the observed text length average of 113.3 tokens, a maximum sequence length of around 128 tokens will provide sufficient context for classification without increasing the risk of overfitting, resulting in better macro F1 performance.

- Evaluation: Not supported. Longer sequences (256 tokens) performed better, confirming the need for sufficient context but exceeding the initial estimate.

Hypothesis 2.1: When hyperparameters are kept consistent, applying data augmentation to class 4 (*Personality Disorder*) and class 5 (*Stress*) will yield a higher macro F1 score and evaluation accuracy than the "Tuned BERT-base" model.

- Evaluation: Partially supported. Augmenting both classes reduced *Personality Disorder* performance but strengthened *Stress* performance. Augmenting only *Stress* improved *Stress* F1 and caused both macro F1 score and accuracy to surpass those of "Tuned BERT-base."

Hypothesis 3.1: MentalBERT's advanced and domain-specific architecture will outperform the "BERT-base baseline" model, primarily in minority class detection.

- Evaluation: Partially supported. MentalBERT slightly improved overall performance; *Stress* F1 increased, but *Personality Disorder* F1 decreased slightly.

Hypothesis 3.2: The data-augmented MentalBERT model will outperform the baseline MentalBERT model when only class 5 (*Stress*) is augmented.

- Evaluation: Supported. Targeted *Stress* augmentation increased macro F1 and *Stress* F1.

6.4 Interpretation of Performance Patterns

The *Normal* class consistently achieved the highest F1 scores, maintaining performance above 96% across all models. Confusion matrix analysis further revealed that the *Normal* class consistently held the highest precision, with very few false positives. This is likely due to polar linguistic distinctions.

Despite being larger classes, *Depression* and *Suicidal* demonstrated persistently lower performance throughout the study. Examination of confusion matrices revealed that a substantial number of true *Suicidal* statements were repeatedly misclassified as *Depression* across every model tested. Neither hyperparameter tuning, augmentation, nor architectural changes meaningfully resolved this confusion, indicating that the challenge stems from inherent linguistic similarity between these categories rather than model optimization alone.

The *Stress* class demonstrated the strongest responsiveness to augmentation, showing dramatic F1 improvements across both architectures following targeted synonym replacement. Personality Disorder, suffered from augmentation, likely due to augmented statements

overpowering the original statements. These findings reinforce the importance of evaluating models using macro F1 rather than accuracy alone, as accuracy can obscure persistent weaknesses in underrepresented classes.

6.5 Study Limitations

Several limitations should be considered when interpreting these results. The hyperparameter search range was constrained by computational resources, and data augmentation was limited to synonym replacement. Synonym replacement's effect was only examined on *Personality Disorder* and *Stress*. Models may even benefit from other augmentation proportions.

The study also relied on a single dataset, limiting generalizability to other platforms or clinical settings. Finally, the persistent confusion between Depression and Suicidal inhibited model performance.

Despite these limitations, the findings demonstrate that coordinated optimization of architecture, training strategy, and data can meaningfully improve performance in transformer-based multi-class mental health text classification.

7. Conclusion

This study compared the effectiveness of hyperparameter tuning, targeted data augmentation, and domain-specific architecture in improving multi-class mental health text classification. The best results emerged from combining all three, with the final optimized MentalBERT model achieving a macro F1 of 86.25% compared to 83.34% in the baseline. Targeted augmentation of the Stress class produced the most substantial gains, while MentalBERT provided additional benefits when paired with tuning and augmentation. The persistent misclassification between Depression and Suicidal remained an unresolved challenge. Future work could explore more advanced augmentation techniques, alternative architectures such as RoBERTa, and testing on real-world clinical data to further improve generalizability and minority class detection

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